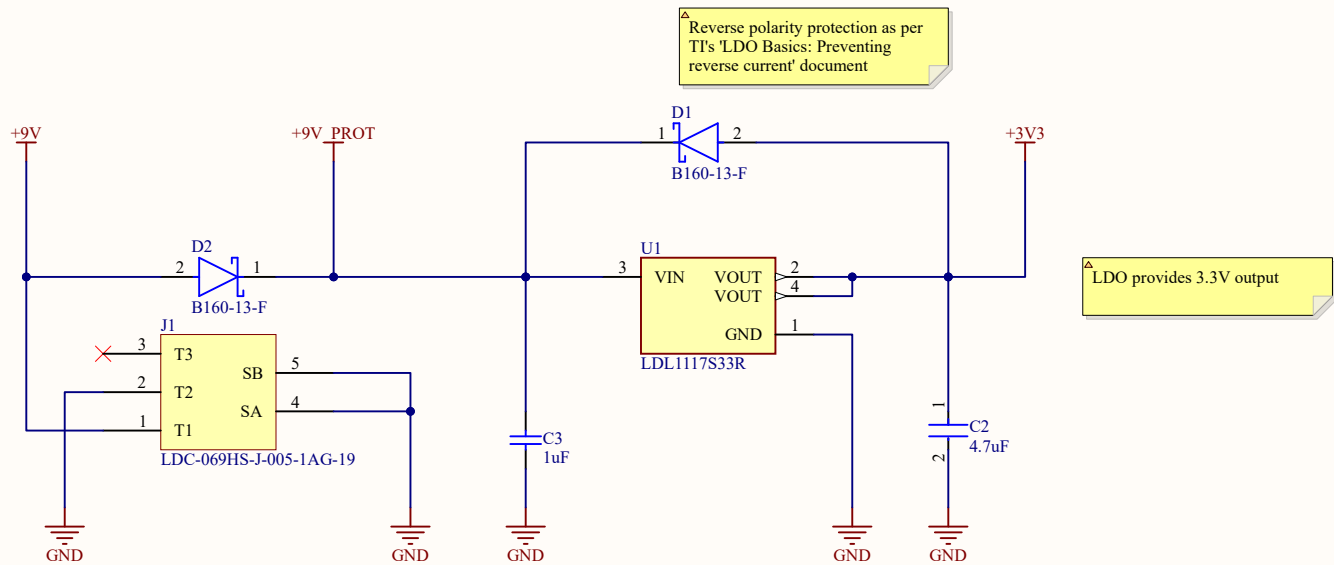
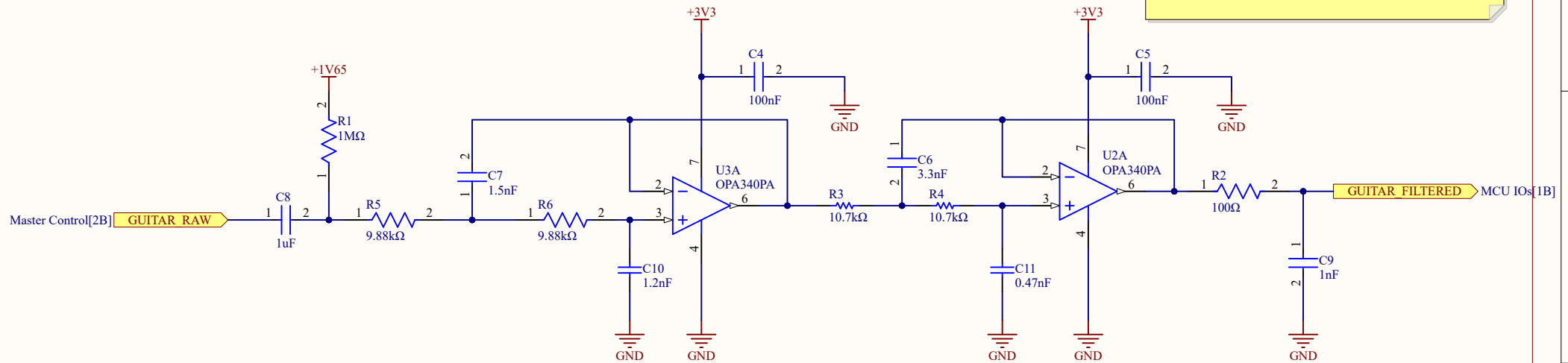




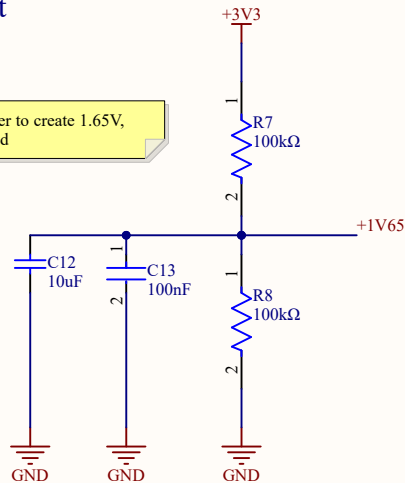
Title		
Board Power		
Size	Number	Revision
A4	1	3
Date:	7/07/2026	Sheet of 7
File:	C:\Users\...\Board Power.SchDoc	Drawn By: Leo Chinchilla

Low-pass Filter



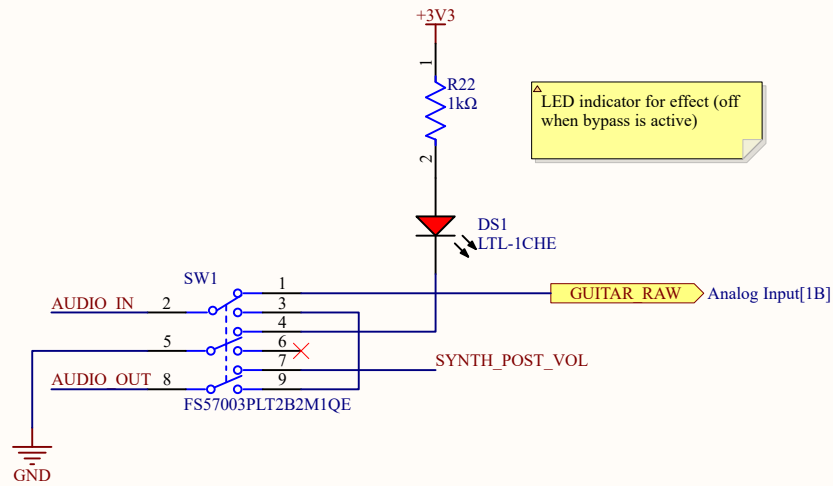
1.65V Net

Voltage divider to create 1.65V, also decoupled

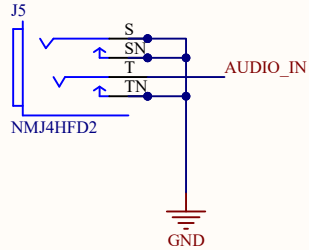


Title		
Analog Input		
Size	Number	Revision
A4	2	3
Date:	7/07/2026	Sheet of 7
File:	C:\Users\...\Analog Input.SchDoc	Drawn By: Leo Chinchilla

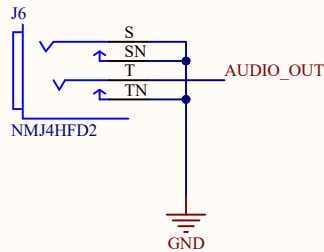
Footswitch



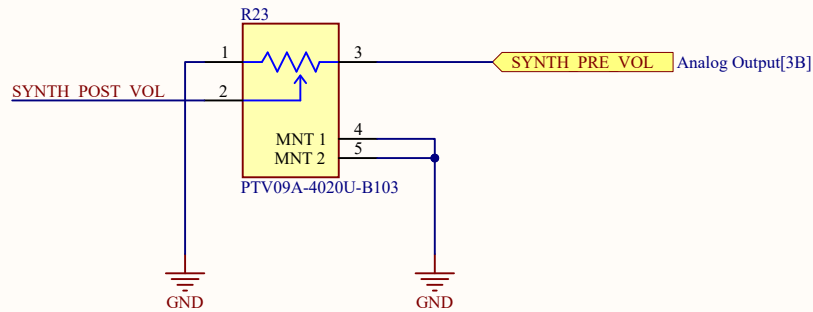
Guitar Input



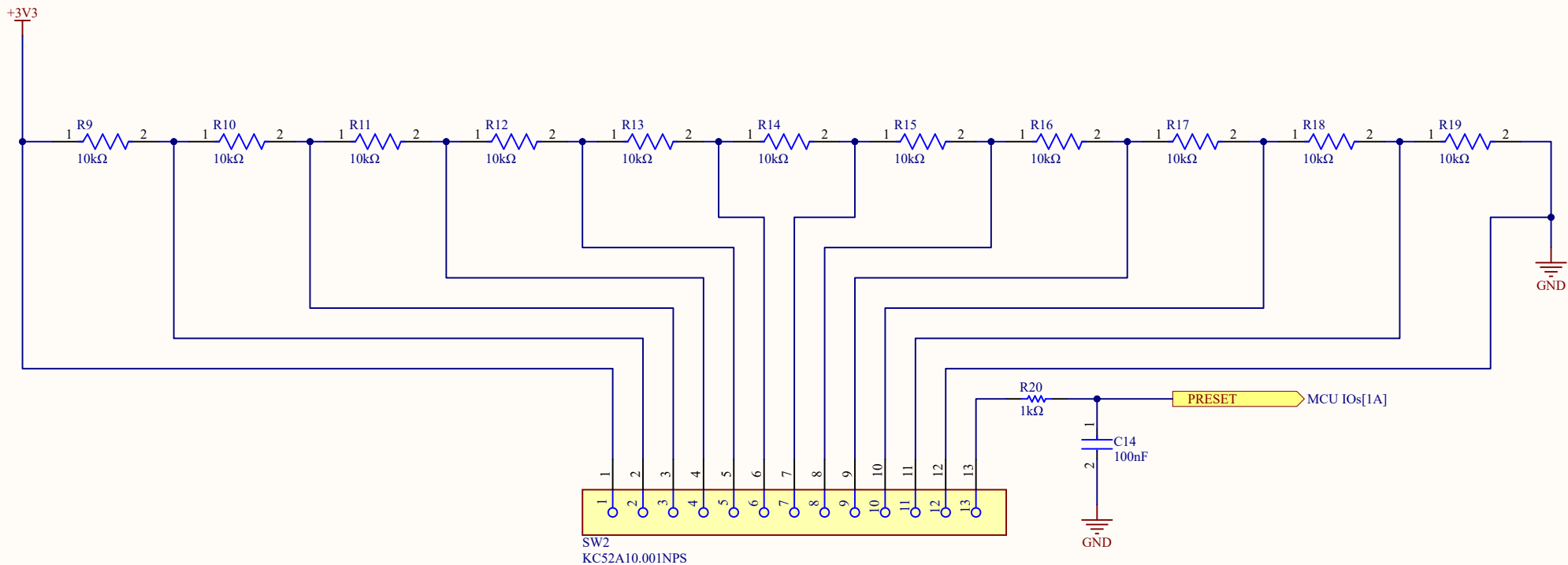
Pedal Output



Potentiometer



Title Master Control		
Size A4	Number 3	Revision 3
Date:	7/07/2026	Sheet of 7
File:	C:\Users\...\Master Control.SchDoc	Drawn By: Leo Chinchilla

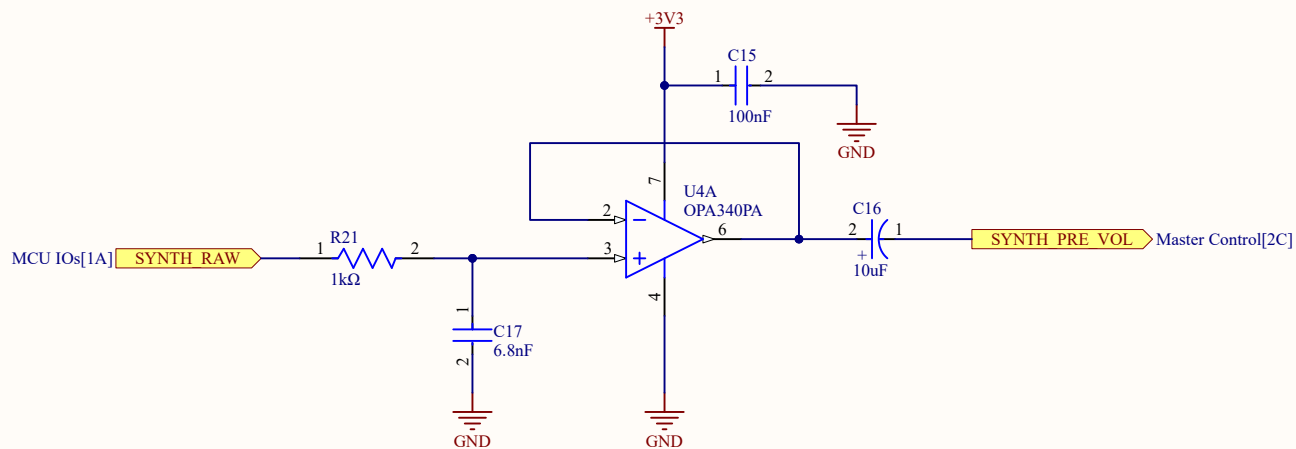


SW2
KC52A10.001NPS

Rotary switch selects 12 different voltage values and passes to the STM32's ADC.

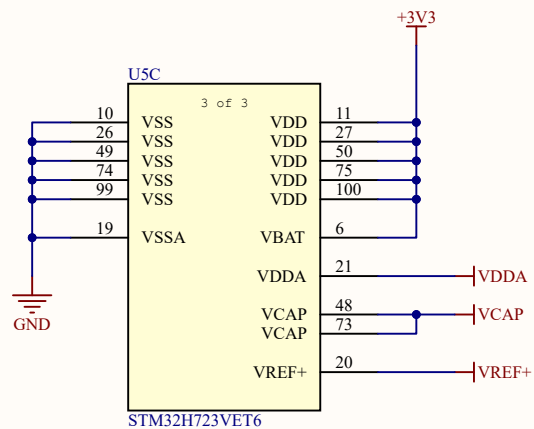
Each voltage gets mapped to a different preset selection.

Title Preset Selection		
Size A4	Number 4	Revision 3
Date:	7/07/2026	Sheet of 7
File:	C:\Users\...\Preset Selection.SchDoc	Drawn By: Leo Chinchilla



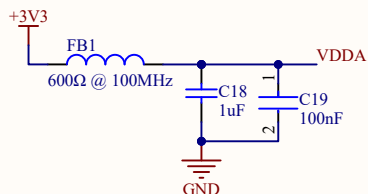
Title			Analog Output	
Size	Number		Revision	
A4	5		3	
Date:	7/07/2026		Sheet of	7
File:	C:\Users\...\Analog Output.SchDoc		Drawn By:	Leo Chinchilla

MCU Power



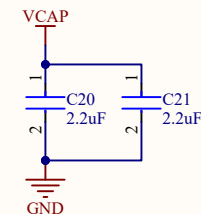
VDDA

△ VDDA decaps as close as possible to the VDDA pin. VDDA connects to VDD through the ferrite bead.



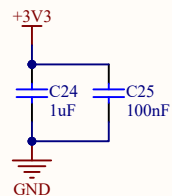
VCAP

△ Placed as close as possible to VCAP pins



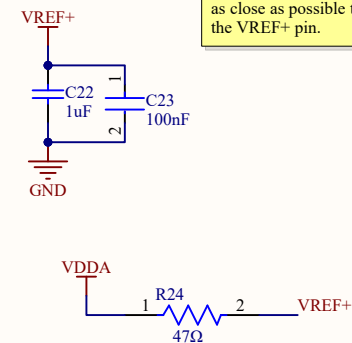
VBAT

△ VBAT decaps to be placed as close as possible to the VBAT pin.



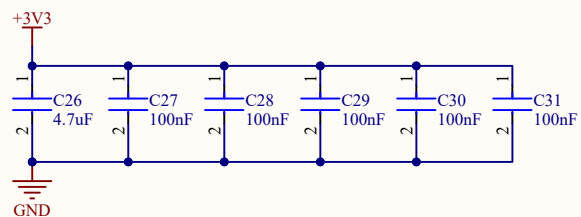
VREF+

△ VREF+ tied to VDDA with a 47ohm resistor, decaps should be placed as close as possible to the VREF+ pin.



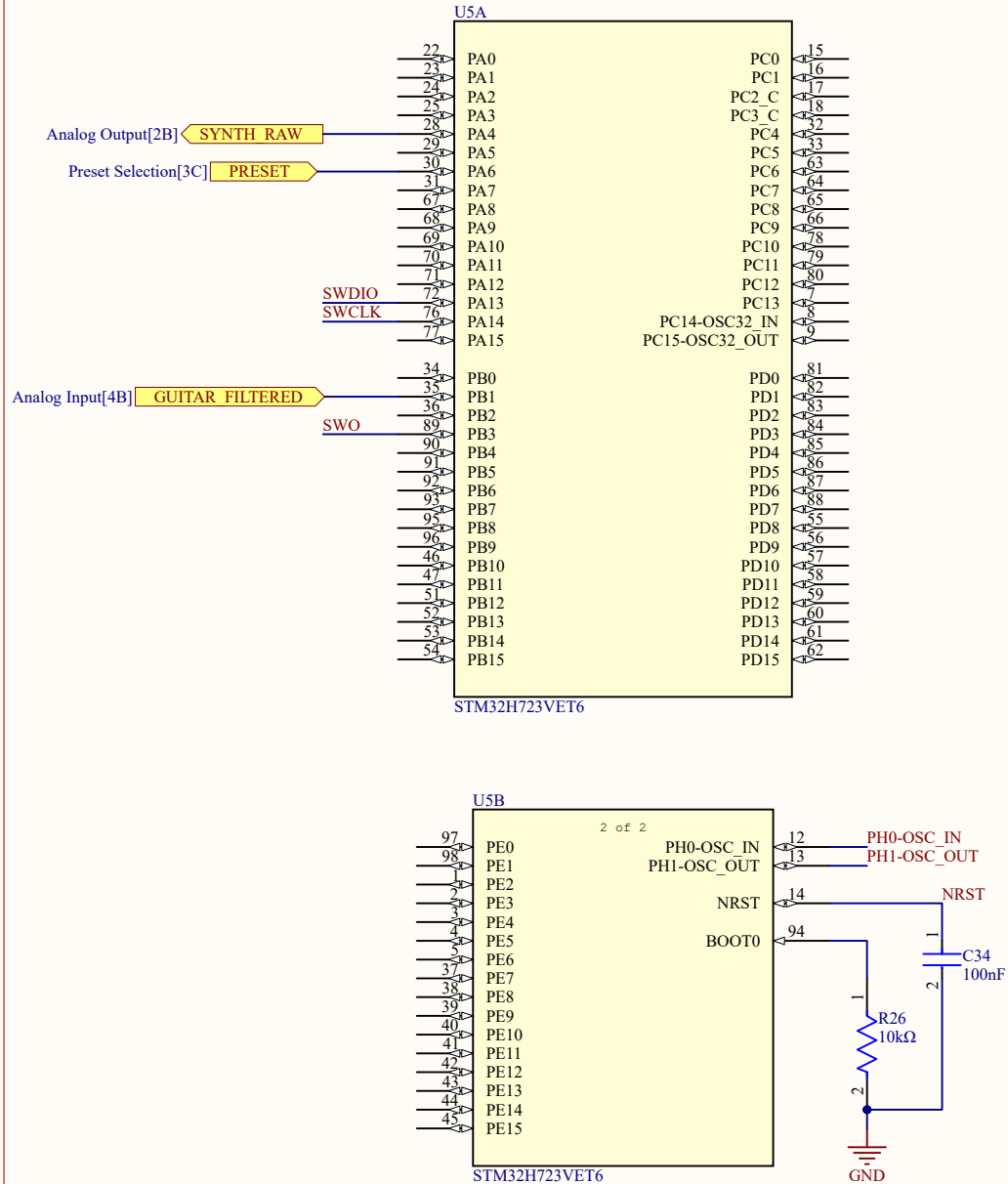
VDD

△ VDD decaps to be placed as close as possible to VDD pins. 4.7uF decap before any one of the VDD pins, pick most suitable.

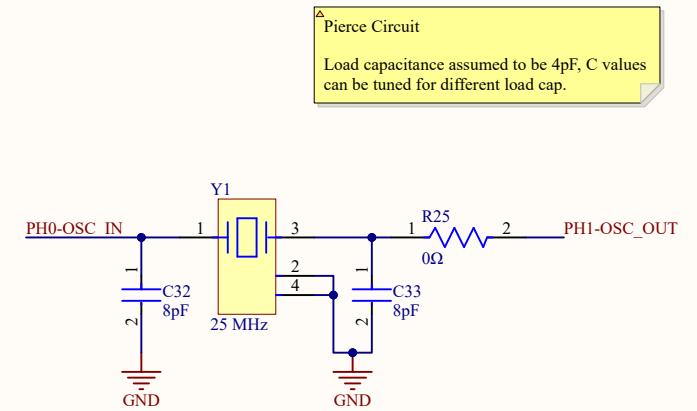


Title MCU Power			
Size A4	Number 6	Revision 3	
Date:	7/07/2026	Sheet of	7
File:	C:\Users\...\MCU Power.SchDoc	Drawn By:	Leo Chinchilla

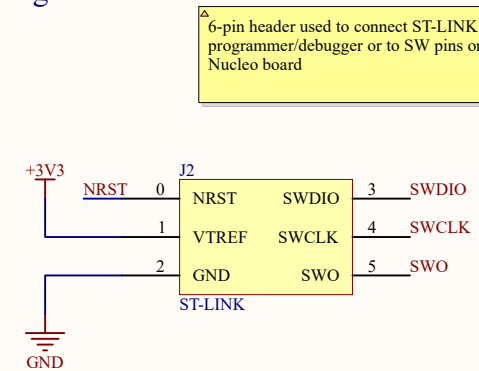
MCU IOs



Crystal Oscillator



SW Debug



Title			MCU IOs	
Size	Number	Revision		
A4	7	3		
Date:	7/07/2026	Sheet of 7		
File:	C:\Users\...\MCU IOs.SchDoc	Drawn By: Leo Chinchilla		